

D875

Secondary Science Teaching Methods

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

Learners build density columns with colorful syrups and oils bought at a grocery store. Everyone is busy and having fun, and the room is loud. When the columns are finished, what should the teacher do next?

- A. Nothing more; hands-on activities always produce learning on their own.
- B. Move on to the next lab to keep the energy up.
- C. Lead a discussion where learners use their observations to explain why the liquids layered in that order.
- D. Grade the columns for neatness and color.

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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2.

Just before lunch, a teacher shows a question on the slide in 32-point font: “Why does ice melt faster on the metal block than on the plastic one?” She calls on the first raised hand after about one second. Only a few learners ever volunteer. What change would most improve participation?

- A. Keep calling on the same volunteers, since they always know the answer.
- B. Wait at least 3-5 seconds before calling on anyone.
- C. Switch to yes-or-no questions so more learners can answer.
- D. Answer the question herself to keep the pace.

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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3. [TRANSFER — reversed stem / different variable]

Research on increasing wait time to 3 seconds or more most consistently reports which outcome?

- A. More off-task behavior every time
- B. Longer, more complete answers from a wider range of learners
- C. Faster lesson pacing
- D. Fewer learners answering

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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4.

In a lab notebook with a purple cover, after a lab that took two class periods, a learner writes: "The mystery liquid is ethanol, because it boiled at 78 °C." Which part of a complete scientific explanation is missing?

- A. Reasoning: that each pure substance has a characteristic boiling point, so a match identifies it
- B. A hypothesis
- C. The claim
- D. The evidence

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Which statement is reasoning rather than a claim or evidence?

- A. The liquid boiled at 78 °C.
- B. The liquid is ethanol.
- C. Each pure substance has its own boiling point, so a matching boiling point identifies the liquid.
- D. Ethanol is used in hand sanitizer.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A teacher who will grade about 150 lab reports this term, usually with a green pen, wants learners to see exactly which parts — claim, evidence, reasoning, and graph — were strong or weak. Which scoring tool fits?

- A. A holistic rubric with one overall score
- B. A completion checklist
- C. A percent-correct score only
- D. An analytic rubric with a row for each part

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A teacher must quickly give one overall score to many short written responses and does not need part-by-part feedback. Which rubric type fits?

- A.** A completion checklist
- B.** No rubric, since short responses never need criteria
- C.** Holistic
- D.** Analytic

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A teacher is writing items for the district’s new online testing platform for a states-of-matter unit. Which prompt is three-dimensional?

- A.** “List the three common states of matter.”
- B.** “Define melting point.”
- C.** “Any question on an online platform is automatically one-dimensional.”
- D.** “This graph shows ice warming and melting. Use particle motion to explain why the temperature stays flat during melting.”

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A learner turned in a lab report two days early, typed in a decorative font. The graph has no units on the x-axis. Which teacher comment best moves learning forward?

- A.** “Great job!”
- B.** “6/10”
- C.** “Graphs are always your weakness.”
- D.** “Your x-axis has no units. Add them so a reader can compare trials, then check your y-axis the same way.”

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

Learners drew models of how a smell from a cinnamon air freshener bought at a dollar store spreads across the room. New data show the smell reaches the back row much faster when a fan is on. What should learners do with their models?

- A. Revise their models to include air movement carrying particles, and note what evidence led to the change.
- B. Copy the teacher's correct model instead.
- C. Keep the original models, because a model is final once drawn.
- D. Replace the model with a written definition of diffusion.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Which learner statement shows a misunderstanding of how models work in science?

- A. "Our model was wrong, so the whole idea of making models is useless."
- B. "Our model needs a new arrow to show the fan's effect."
- C. "Every model leaves some things out."
- D. "We should test whether the model predicts the next trial."

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A lab uses a 12-step procedure printed on both sides of one page. Several learners struggle to follow the reading and fall behind. What is the best differentiation?

- A. Excuse them from the lab.
- B. Keep the same lab goal and add picture step cards and a partner reader.
- C. Give those learners a coloring worksheet on lab equipment instead.
- D. Give every learner identical materials, since fairness always means sameness.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

Learners build density columns ~~with colorful syrups and oils bought at a grocery store~~. Everyone is busy and having fun, ~~and the room is loud~~. When the columns are finished, what should the teacher do next?

Target word: next step after a hands-on activity **Decides:** Sensemaking must follow the activity.

A	Absolute language	Activity without reflection often leaves misconceptions intact.
B	Dropped element	Skips sensemaking, so the activity may not produce learning.
C	Correct	Turns the activity into evidence-based explanation.
D	Adjacent category	Scores the product, not understanding.

2. Answer B Concept 2

~~Just before lunch~~, a teacher shows a question ~~on the slide in 32-point font~~: “Why does ice melt faster on the metal block than on the plastic one?” She calls on the first raised hand after about one second. Only a few learners ever volunteer. What change would most improve participation?

Target word: questioning move **Decides:** One second is too short; wait time is the fix.

A	Absolute language	Leaves most learners out of the thinking.
B	Correct	More think time brings more and better answers.
C	Adjacent category	Closed questions lower the thinking demand.
D	Inverted operation	Removes thinking from learners entirely.

3. Answer B Concept 2 · transfer

Research on increasing wait time to 3 seconds or more most consistently reports which outcome?

Target word: effect of wait time **Decides:** Wait time raises answer quality and participation.

A	Absolute language	Not a consistent finding.
B	Correct	Think time helps more learners form full responses.
C	Confusion set	Wait time slows pace slightly; its payoff is quality.
D	Inverted operation	The opposite of the typical finding.

4. Answer A Concept 3

~~In a lab notebook with a purple cover~~, after a lab ~~that took two class periods~~, a learner writes: “The mystery liquid is ethanol, because it boiled at 78 °C.” Which part of a complete scientific explanation is missing?

Target word: missing CER element **Decides:** Claim and evidence are present; the principle linking them is not.

A	Correct	Reasoning explains why the evidence supports the claim.
B	Adjacent category	A hypothesis comes before testing, not in the final argument.
C	Inverted operation	“The liquid is ethanol” is the claim, and it is present.
D	Confusion set	The 78 °C measurement is the evidence.

5. Answer C Concept 3 • transfer

Which statement is reasoning rather than a claim or evidence?

Target word: identify reasoning **Decides:** Reasoning applies a science principle.

A	Confusion set	That is evidence.
B	Adjacent category	That is the claim.
C	Correct	States the principle that links data to claim.
D	Camouflage	A true fact that does not connect evidence to claim.

6. Answer D Concept 4

A teacher who will grade ~~about 150~~ lab reports this term, ~~usually with a green pen,~~ wants learners to see exactly which parts — claim, evidence, reasoning, and graph — were strong or weak. Which scoring tool fits?

Target word: rubric type for part-by-part feedback **Decides:** Separate criteria need separate scores.

A	Confusion set	Fast, but does not show which part was weak.
B	Adjacent category	Shows whether parts exist, not their quality.
C	Dropped element	Leaves out any criterion-level feedback.
D	Correct	Scores each criterion and shows where to improve.

7. Answer C Concept 4 • transfer

A teacher must quickly give one overall score to many short written responses and does not need part-by-part feedback. Which rubric type fits?

Target word: rubric type for a quick overall judgment **Decides:** One overall score, speed matters.

A	Adjacent category	Checks presence, not quality.
B	Absolute language	Criteria keep even quick scoring consistent.
C	Correct	One score for the whole response, efficient for large volumes.
D	Inverted operation	Gives part-by-part scores, which is slower and not needed here.

8. Answer D Concept 5

A teacher is writing items ~~for the district's new online testing platform~~ for a states-of-matter unit. Which prompt is three-dimensional?

Target word: identify a three-dimensional task **Decides:** It requires a practice, a crosscutting concept, and a core idea applied to a phenomenon.

A	Dropped element	No practice or crosscutting concept.
B	Adjacent category	Recall of one idea only.
C	Absolute language	Format does not decide dimensionality; the task demand does.
D	Correct	Analyzing data (practice), energy and matter (crosscutting), particle model (core idea).

9. Answer D Concept 6

A learner turned in a lab report ~~two days early, typed in a decorative font~~. The graph has no units on the x-axis. Which teacher comment best moves learning forward?

Target word: effective feedback **Decides:** Specific, tied to criteria, with a next step.

A	Dropped element	Praise without information.
B	Adjacent category	A score alone does not tell the learner what to fix.
C	Absolute language	Labels the learner instead of the work.
D	Correct	Names the gap, the reason, and the next step.

10. Answer A Concept 7

Learners drew models of how a smell ~~from a cinnamon air freshener bought at a dollar store~~ spreads across the room. New data show the smell reaches the back row much faster when a fan is on. What should learners do with their models?

Target word: use of new evidence with models **Decides:** Evidence that does not fit calls for revision.

A	Correct	Models are revised to account for new evidence.
B	Dropped element	Skips the learners' own reasoning about evidence.
C	Absolute language	Scientific models are tentative.
D	Adjacent category	A definition does not explain the new pattern.

11. Answer A Concept 7 • transfer

Which learner statement shows a misunderstanding of how models work in science?

Target word: spot the misconception **Decides:** Models change with evidence.

A	Correct	Treats revision as failure instead of how science progresses.
B	Inverted operation	That is correct model revision.
C	Confusion set	Accurate: all models are simplifications.
D	Adjacent category	Accurate use of a model to predict.

12. Answer B Concept 8

A lab uses a 12-step procedure ~~printed on both sides of one page~~. Several learners struggle to follow the reading and fall behind. What is the best differentiation?

Target word: appropriate differentiation **Decides:** Keep the goal; change the support.

A	Dropped element	Removes access to the learning.
B	Correct	Scaffolds access without lowering expectations.
C	Adjacent category	Changes to an easier goal.
D	Absolute language	Equity means access to the same goal, not identical supports.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	11	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	5	
Absolute language	Always, never, guarantee — where judgment applies.	8	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Hands-on needs minds-on	A hands-on activity becomes learning only when learners make sense of what they observed through discussion and explanation.	Assuming a fun, busy activity means learning happened.	A lab ends and the teacher decides what comes next.
2	Wait time	Waiting 3 or more seconds after a question (and after an answer) leads to longer, better answers from more learners.	Calling on the first hand to keep pace.	A question followed by a quick call or silence.
3	Claim, evidence, reasoning	A scientific argument states a claim, supports it with data, and uses a science principle to explain why the data support the claim.	Treating a data point as the reasoning.	A learner explanation with one part missing.
4	Analytic vs. holistic rubrics	Analytic rubrics score each criterion separately and give targeted feedback; holistic rubrics give one overall score quickly.	Choosing a holistic rubric when learners need part-by-part feedback.	The purpose of scoring: feedback on parts vs. one quick judgment.
5	Three-dimensional assessment tasks	A three-dimensional task asks learners to use a practice and a crosscutting concept with a core idea to explain a phenomenon or solve a problem.	Counting a recall or definition question as three-dimensional.	Several assessment prompts compared.
6	Feedback that moves learning	Effective feedback is specific, tied to the criteria, and tells the learner what to do next.	Giving praise or a score alone.	Candidate comments on a learner's work.
7	Revising models with evidence	Scientific models are tentative and are revised when new evidence does not fit.	Treating a first model as a final answer, or replacing learner models with the teacher's.	New data conflict with a learner-drawn model.
8	Scaffold, don't lower the goal	Differentiation keeps the same learning goal and changes supports such as visuals, sentence frames, or grouping.	Giving struggling learners an easier goal or excusing them from the task.	Some learners struggle with the reading or steps in a lab.

Course scope (public description): This course focuses on equipping secondary science educators with the essential knowledge and skills to effectively teach science through a three-dimensional approach, integrating science and engineering practices, crosscutting concepts, and disciplinary core ideas. Participants will explore general considerations for science instruction, including inquiry-based learning, hands-on activities, and assessment strategies. By building on foundational knowledge in Three-Dimensional Science and Engineering, educators will enhance their ability to engage students in meaningful and authentic scientific learning experiences. This methods course will count towards 7 hours of the 51 total clinical hours that learners will gain from their SCED methods courses.